PAPER_940: TXS 0506+056 Jet Power Curves

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** blazar_jet_power_curves_extended.py (TXS0506JetPowerCurves) **Calculator:** TXS0506JetPowerCurvesCalc (CP4 #524) **CVW:** v2.0.0 compliant

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Abstract

We derive UQFF phonon-modulated jet power curves for TXS 0506+056, the blazar associated with the IceCube-170922A high-energy neutrino event. With $M_{\rm BH} = 3 \times 10^8\, M_\odot$, $a = 0.85$, $B = 8000$ T, and $A_{\rm jet} = 1.20$, the Blandford-Znajek power is enhanced by $2.9\times/2.3\times/1.6\times$ at $\Gamma = 0.05/0.10/0.30$ THz. The elevated spin and magnetic field reflect the extreme jet conditions required for neutrino production.

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1. System Parameters

Parameter	Value	Source
$M_{\rm BH}$	$3 \times 10^8\, M_\odot$	Spectral modeling
Spin a	0.85	Jet Lorentz factor
B	8000 T	Synchrotron SED
$A_{\rm jet}$	1.20	Phonon coupling
Redshift z	0.3365	SDSS
Neutrino	IceCube-170922A	IceCube 2018

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2. Jet Power Curves

$$P_{\rm jet}(\Gamma) = P_{\rm BZ} \cdot \left(1 + M_{\rm jet}(\Gamma)\right)$$

Γ (THz)	Enhancement	Regime
0.05	$2.9\times$	Narrow
0.10	$2.3\times$	Optimal
0.30	$1.6\times$	Broad

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3. IceCube Association

The high $A_{\rm jet} = 1.20$ coupling strength implies that phonon-jet modulation produces sufficient hadronic acceleration for ~ 290 TeV neutrino production via $p\gamma$ interactions, consistent with IceCube-170922A timing.

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4. Source Data

- **File:** blazar_jet_power_curves_extended.py
- **Session:** 213
- **CP4 Class:** TXS0506JetPowerCurvesCalc (#524)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. IceCube Collaboration (2018) – Science, 361, eaat1378
3. Padovani, P. et al. (2018) – MNRAS, 480, 192